

CHAPTER 3

Chemical Kinetics

Rate of Chemical Reactions

1. For a reaction $3A \rightarrow 2B$

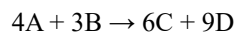
The average rate of appearance of B is given by $\frac{\Delta[B]}{\Delta t}$. The correct relation between the average rate of appearance of B with the average rate of disappearance of A is given in option: (2023-Manipur)

- a. $\frac{-\Delta[A]}{\Delta t}$ b. $\frac{-3\Delta[A]}{2\Delta t}$ c. $\frac{-2\Delta[A]}{3\Delta t}$ d. $\frac{\Delta[A]}{\Delta t}$

2. The correct options for the rate law that corresponds to overall first order reaction is (2023-Manipur)

- a. Rate = $k[A]^0 [B]^2$ b. Rate = $k[A] [B]$
c. Rate = $k[A]^{1/2} [B]^2$ d. Rate = $k[A]^{-1/2} [B]^{3/2}$

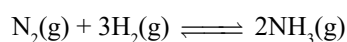
3. For a chemical reaction (2022 Re)



rate of formation of C is $6 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and rate of disappearance of A is $4 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of reaction and amount of B consumed in interval of 10 seconds, respectively will be:

- a. $10 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $30 \times 10^{-2} \text{ mol L}^{-1}$
b. $1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $30 \times 10^{-2} \text{ mol L}^{-1}$
c. $10 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $10 \times 10^{-2} \text{ mol L}^{-1}$
d. $1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $10 \times 10^{-2} \text{ mol L}^{-1}$

4. For the chemical reaction



The correct option is: (2019)

- a. $-\frac{1}{3} \frac{d[H_2]}{dt} = -\frac{1}{2} \frac{d[NH_3]}{dt}$ b. $-\frac{d[N_2]}{dt} = 2 \frac{d[NH_3]}{dt}$
c. $-\frac{d[N_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt}$ d. $3 \frac{d[H_2]}{dt} = 2 \frac{d[NH_3]}{dt}$

Factors Influencing Rate of a Reaction & Order of Reaction

5. Following data is for a reaction between reactants A and B:

Rate $\text{mol L}^{-1} \text{ s}^{-1}$	[A]	[B]
2×10^{-3}	0.1 M	0.1 M
4×10^{-3}	0.2 M	0.1 M
1.6×10^{-2}	0.2 M	0.2 M

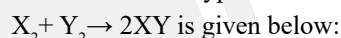
The order of the reaction with respect to A and B, respectively, are (2024 Re)

- a. 1,0 b. 0,1 c. 1,2 d. 2,1

6. For a certain reaction, the rate = $k[A]^2[B]$. When the initial concentration of A is tripled keeping concentration of B constant, the initial rate would (2023)

- a. increase by a factor of three.
b. decrease by a factor of nine.
c. increase by a factor of six.
d. increase by a factor of nine

7. Mechanism of a hypothetical reaction (2017-Delhi)



- (i) $X_2 \rightarrow X + X$ (fast) (ii) $X + Y_2 \rightleftharpoons XY + Y$ (slow)
(iii) $X + Y \rightarrow XY$ (fast)

The overall order of the reaction will be

- a. 1.5 b. 1 c. 2 d. 0

8. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first-order reaction. It is because the: (2016-II)

- a. Rate is proportional to the surface coverage
b. Rate is inversely proportional to the surface coverage
c. Rate is independent of the surface coverage
d. Rate of decomposition is very low

Integrated Rate Equation & Half Life of Reactions

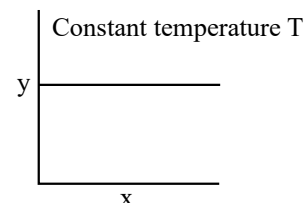
9. The half life of a first order reaction is 2000 years. If the concentration after 8000 years is 0.02 M, then the initial concentration was: (2022 Re)

- a. 0.04 M b. 0.16 M c. 0.32 M d. 0.08 M

10. For a first order reaction $A \rightarrow \text{Products}$, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min^{-1} is (2022)

- a. 0.2303 b. 1.3818 c. 0.9212 d. 0.4606

11. The given graph is a representation of kinetics of a reaction.



The y and x axes for zero and first order reactions, respectively are (2022)

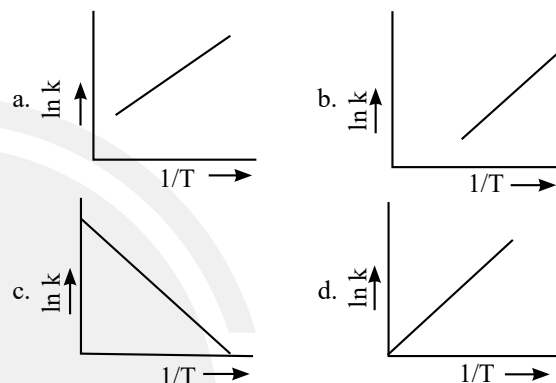
- a. zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = \text{rate}$ and $x = t_{1/2}$)
 b. zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = t_{1/2}$ and $x = \text{concentration}$)
 c. zero order ($y = \text{concentration}$ and $x = \text{time}$), first order ($y = \text{rate constant}$ and $x = \text{concentration}$)
 d. zero order ($y = \text{rate}$ and $x = \text{concentration}$), first order ($y = t_{1/2}$ and $x = \text{concentration}$)
12. The rate constant for a first order reaction is $4.606 \times 10^{-3} \text{ s}^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is: (2020)
 a. 200 s b. 500 s c. 1000 s d. 100 s
13. The half-life for a zero order reaction having 0.02 M initial concentration of reactant is 100 s. The rate constant (in $\text{mol L}^{-1} \text{ s}^{-1}$) for the reaction is (2020-Covid)
 a. 2.0×10^{-4} b. 2.0×10^{-3}
 c. 1.0×10^{-2} d. 1.0×10^{-4}
14. If the rate constant for a first order reaction is k , the time (t) required for the completion of 99% of the reaction is given by: (2019)
 a. $t = 0.693/k$ b. $t = 6.909/k$
 c. $t = 4.606/k$ d. $t = 2.303/k$
15. When initial concentration of the reactant is doubled, the half-life period of a zero order reaction (2018)
 a. Is halved b. Is doubled
 c. Is tripled d. Remains unchanged
16. The correct difference between first and second order reactions is that: (2018)
 a. The rate of a first-order reaction does not depend on reactant concentrations; the rate of a second-order reaction does depend on reactant concentrations
 b. The half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction does depend on $[A]_0$
 c. The rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations
 d. A first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed
17. A first order reaction has a specific reaction rate of 10^{-2} s^{-1} . How much time will it take for 20 g of the reactant to reduce to 5 g? (2017-Delhi)
 a. 693.0 second b. 238.6 second
 c. 138.6 second d. 346.5 second
18. The rate of a first-order reaction is $0.04 \text{ mol L}^{-1} \text{ s}^{-1}$ at 10 seconds and $0.03 \text{ mol L}^{-1} \text{ s}^{-1}$ at 20 seconds after initiation of the reaction. The half-life period of the reaction is: (2016 - I)
 a. 54.1 s b. 24.1 s c. 34.1 s d. 44.1 s
19. The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} \text{ mole per second}$. If the concentration of A is 5 M, then concentration of B after 20 minutes is: (2015 Re)
 a. 0.72 M b. 1.08 M c. 3.60 M d. 0.36 M

20. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is: (2015)

- a. First
 b. Second
 c. More than zero but less than first
 d. Zero

Temperature Dependence of the Rate of a Reaction & Effect of Catalysts

21. Which of the following plot represents the variation of $\ln k$ versus $\frac{1}{T}$ in accordance with Arrhenius equation? (2024 Re)



22. Rate constants of a reaction at 500 K and 700 K are 0.04 s^{-1} and 0.14 s^{-1} , respectively; then, activation energy of the reaction is: (2024 Re)

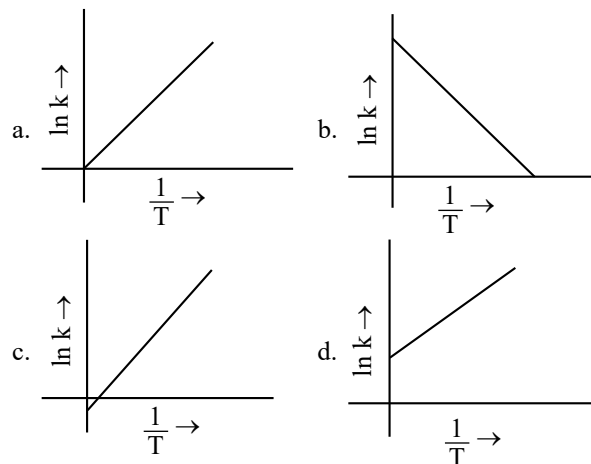
(Given: $\log 3.5 = 0.5441$, $R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$)

- a. 182310 J b. 18500 J c. 18219 J d. 18030 J

23. Activation energy of any chemical reaction can be calculated if one knows the value of (2024)

- a. orientation of reactant molecules during collision
 b. rate constant at two different temperatures
 c. rate constant at standard temperature
 d. probability of collision

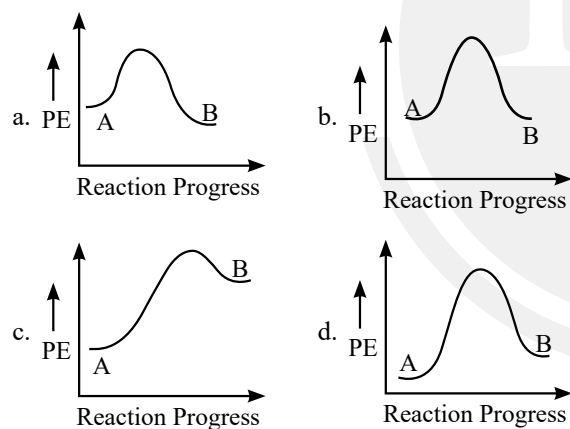
24. Which plot of $\ln k$ vs $\frac{1}{T}$ is consistent with Arrhenius equation? (2024)



25. The rate of a reaction quadruples when temperature changes from 27°C to 57°C . Calculate the energy of activation. (2024)

Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 4 = 0.6021$

- a. 3.80 kJ/mol b. 3804 kJ/mol
c. 38.04 kJ/mol d. 380.4 kJ/mol
26. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R: (2023)
- Assertion A:** A reaction can have zero activation energy.
- Reason R:** The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.
- In the light of the above statements, choose the correct answer from the options given below:
- a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.
c. Both A and R are true and R is NOT the correct explanation of A.
d. A is true but R is false.
27. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option. (2021)



28. The slope of Arrhenius Plot ($\ln K$ v/s $1/T$) of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is. Choose the correct option for your answer. (2021)
- [Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]
- a. 83.0 kJ mol^{-1} b. 166 kJ mol^{-1}
c. -83 kJ mol^{-1} d. 41.5 kJ mol^{-1}
29. The addition of a catalyst during a chemical reaction alters which of the following quantities? (2016 - I)
- a. Activation energy b. Entropy
c. Internal energy d. Enthalpy
30. The activation energy of a reaction can be determined from the slope of which of the following graphs? (2015)
- a. $\frac{\ln k}{T}$ vs. T b. $\ln k$ vs. $\frac{1}{T}$ c. $\frac{T}{\ln k}$ vs. $\frac{1}{T}$ d. $\ln k$ vs. T
31. A reaction having equal energies of activation for forward and reverse reactions has: (2013)
- a. $\Delta S = 0$ b. $\Delta G = 0$
c. $\Delta H = 0$ d. $\Delta H = \Delta G = \Delta S = 0$
32. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 20°C to 35°C ? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$) (2013)
- a. 342 kJ mol^{-1} b. 269 kJ mol^{-1}
c. 34.7 kJ mol^{-1} d. 15.1 kJ mol^{-1}

Collision Theory of Chemical Reactions

33. In collision theory of chemical reaction, Z_{AB} represents (2020-Covid)
- a. The collision frequency of reactants, A and B
b. Steric factor
c. The fraction of molecules with energies equal to E_a
d. The fraction of molecules with energies greater than E_a
34. An increase in the concentration of the reactants of a reaction leads to change in: (2020)
- a. Heat of reaction b. Threshold energy
c. Collision frequency d. Activation energy

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (b) | 4. (c) | 5. (c) | 6. (d) | 7. (a) | 8. (a) | 9. (c) | 10. (c) |
| 11. (d) | 12. (b) | 13. (d) | 14. (c) | 15. (b) | 16. (b) | 17. (c) | 18. (b) | 19. (a) | 20. (a) |
| 21. (c) | 22. (c) | 23. (b) | 24. (b) | 25. (c) | 26. (c) | 27. (a) | 28. (d) | 29. (a) | 30. (b) |
| 31. (c) | 32. (c) | 33. (a) | 34. (c) | | | | | | |